

KEYWORDS

bite force, ecology, evolution, performance, pinching force, zoology

1 | INTRODUCTION

Closing forces, such as bite and pinching forces, are important performance metrics for many animals. They can relate to dietary spectra (e.g. Aguirre et al., 2002; Meyers et al., 2018; Shi et al., 2020), influence the outcome of conflicts (e.g. Claussen et al., 2007; Husak et al., 2006; Lailvaux et al., 2004) and impact reproductive success (e.g. Husak et al., 2009; Lappin & Husak, 2005). The dependencies of closing forces on size, ecology, age or phylogeny have been studied in vivo in a wide range of vertebrate groups, including cartilaginous and bony fishes (Grubich et al., 2012; Huber et al., 2005), crocodilians (e.g. Erickson et al., 2012), birds (e.g. Herrel et al., 2005), turtles (e.g. Herrel et al., 2002), squamates (e.g. Herrel et al., 2001; Jones et al., 2020), frogs (Lappin et al., 2017) and mammals (e.g. Becerra et al., 2011; Dessem & Druzinsky, 1992; Freeman & Lemen, 2008; Sakamoto et al., 2010; Santana & Dumont, 2009). However, just a few studies investigated closing forces in invertebrates so far. We could find published closing force data on seven species of scorpions (van der Meijden et al., 2010), one camel spider (van der Meijden et al., 2012), 19 crustaceans (Block & Rebach, 1998; Blundon, 1988; Bywater et al., 2008; Claussen et al., 2007; Elnor & Campbell, 1981; Govind & Blundon, 1985; Lailvaux et al., 2009; Levinton & Judge, 1993; Malavé et al., 2018; McLain et al., 2015; Miranda et al., 2016; Oka et al., 2016; Smith & Palmer, 1994; South et al., 2020; Taylor, 2000; Taylor et al., 2000; Wilson et al., 2007; Yamada & Boulding, 1998), and 33 insects (David et al., 2016a, 2016b; Hao et al., 2018; Huang, 2012; Paul & Gronenberg, 2002; Spagna et al., 2008; Weihmann et al., 2015; Wheeler & Evans, 1989). The investigated species represent only a small fraction of the vast taxonomic and anatomical diversity of the megadiverse invertebrates. Given their immense importance to ecosystems (Prather et al., 2013) and their potential for understanding the evolution of performance traits, the paucity of closing force data in these taxa is a knowledge gap that warrants attention.

1.1 | Previous closing force measurement methods

Many closing force experiments have relied on custom-made strain gauge-based measurement setups (Claussen et al., 2007; DeChow & Carlson, 1983; Dessem & Druzinsky, 1992; Erickson et al., 2003; Grubich et al., 2012; Hylander, 1979; Paul & Gronenberg, 2002; Weihmann et al., 2015; Williams et al., 2009). The minimum gape distances of these setups are limited by the minimal thickness of the tip elements at which forces are measured, and the distance between the tip elements after their elastic deformation at maximum closing force. Other closing force studies used custom implementations of a setup developed by Herrel et al. (1999) for bite force measurements

in lizards. The setup is based on a piezoelectric force transducer, which, in contrast to strain gauge-based transducers, effectively does not deform under the application of forces. This isometric sensor behaviour has the advantage that the tip elements of the setup do not move towards each other, allowing for gape distances that are force-independent and only limited by the thickness of the tip elements and their distance from each other. The setup was used to measure closing forces in a wide range of animals, for example, a camel spider (van der Meijden et al., 2012), scorpions (van der Meijden et al., 2010), stag beetles (Goyens et al., 2014), lizards (e.g. Lappin et al., 2006), turtles (e.g. Herrel et al., 2018), frogs (Lappin et al., 2017), birds (e.g. Herrel et al., 2005) and bats (e.g. Aguirre et al., 2002).

Animal closing forces were shown to be gape-angle dependent in cockroaches and bats (Dumont & Herrel, 2003; Ross & Iriarte-Diaz, 2014; Santana, 2016; Weihmann et al., 2015; Williams et al., 2009), so the size of and distance between the tip elements of a measurement setup may limit the taxa suited for physiologically meaningful force measurements. While the distance between the tip elements in the Herrel et al. (1999) setup is adjustable, allowing closing force measurements with different gape angles and specimen sizes, the replaceable tip elements of the setup have a minimum thickness of ~2 mm each, resulting in a theoretical minimum gape distance of 4 mm. To our knowledge, so far the smallest species measured with this setup was *Anolis humilis* Peters, 1863 (Squamata: Dactyloidae) with a snout-vent length of 32.7 mm (Wittorski et al., 2016). This is more than three times larger than, for example, the peak of the size-frequency distribution of the megadiverse insects, which lies at ~10 mm (Finlay et al., 2006). The relatively large cockroach *Periplaneta americana* (L. 1758; Hexapoda: Blattodea), which is the only insect species where the gape angle-dependency of bite force was studied, exhibited the strongest bite forces at a mandible gape distance of only 0.5 mm (Weihmann et al. (2015) and Figure S1a). A gape distance of 4 mm, as in the Herrel et al. (1999) setup, would therefore not allow bite force measurements for the majority of insects (Figure S1b). Although the system by Weihmann et al. (2015) allows measuring bite forces at very small gape distances, the operator has to sedate and fix the test animals to the setup with dental cement.

To overcome the size limitations of previous setups, we propose the FORCEX (force eXperiment) measurement setup, a battery-powered, portable system that allows rapid closing force measurements in a wide range of animals, particularly of small invertebrates. FORCEX measurements do not harm the test animals, allowing repeated measurements and a subsequent release, facilitating population level or multi-generation studies. While many previous studies focused only on maximum closing forces (but see Boulding, 1984; Huber et al., 2005; Goyens et al., 2014; Weihmann et al., 2015; David et al., 2016a, 2016b; Gomes et al., 2020), FORCEX is complemented with the new FORCEr package to run in-depth